

CEE 406 Pavement Design I
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Example AASHTO Flexible Pavement Design Method

- Design Period = 20 years (Initial flexible pavement = 15 years)
- 2-way 18-kip ESALs first year = 2.5×10^6
- Projected growth = 3%
- 3-lane in one direction
- Heavily traffic state rural highway
- The project is 16 miles (80 bore-holes obtained; the first 12 bore-holes were different from the rest)
- The project is in a climate region II (wet-freeze-thaw cycles)
- Soil is highly active swelling clay
- Drainage system is needed to remove water within one day (assume exposure to moisture is 1-5%)
- Depth to rock is 50 ft
- Soils' PI varies and moisture content is at optimum
- Road soil condition is as follows

Duration (months)	Condition	Resilient Modulus (M_r) (psi)
5.0	Wet	5000
5.5	Dry	6500
0.5	Spring thaw	4000
1.0	Frozen	20000

Assume the following:

- $P_o = 4.6$
- HMA resilient modulus = 400,000 psi
- Base resilient modulus = 30,000 psi
- Subbase resilient modulus = 11,000 psi

Bore Hole	Section Length	Roadbed Thickness	PI	Moisture condition	Potential Vertical Rise (in)		Soil Fabric	Swell Rate Constant	
1	900	>30	48	Optimum					
2	1200	>30	56	Optimum					
3	800	>30	67	Optimum					
4	1000	>30	15	Optimum					
5	1000	>30	46	Optimum					
6	1100	>30	61	Optimum					
7	1000	>30	65	Optimum					
8	900	>30	71	Optimum					
9	1200	>30	38	Optimum					
10	800	>30	60	Optimum					
11	900	>30	19	Optimum					
12	1200	>30	50	Optimum					

Swell rate constant: If $PI > 20$, then $\theta = 0.07$

If $PI \leq 20$, then $\theta = 0.1$

Assume relatively tight soil fabric

No potential to swell if $PI < 20$